In [6]:
```
import pandas as pd

# Load Titanic dataset
df = pd.read_csv('tested.csv')
df.head()
```

Out[6]:

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fa |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 892 | 0 | 3 | Kelly, Mr. James | male | 34.5 | 0 | 0 | 330911 | 7.82 |
| **1** | 893 | 1 | 3 | Wilkes, Mrs. James (Ellen Needs) | female | 47.0 | 1 | 0 | 363272 | 7.00 |
| **2** | 894 | 0 | 2 | Myles, Mr. Thomas Francis | male | 62.0 | 0 | 0 | 240276 | 9.68 |
| **3** | 895 | 0 | 3 | Wirz, Mr. Albert | male | 27.0 | 0 | 0 | 315154 | 8.66 |
| **4** | 896 | 1 | 3 | Hirvonen, Mrs. Alexander (Helga E Lindqvist) | female | 22.0 | 1 | 1 | 3101298 | 12.28 |

In [7]:
```
df.info()

df.describe()

df['Survived'].value_counts()
df['Sex'].value_counts()
df['Pclass'].value_counts()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  418 non-null    int64  
 1   Survived     418 non-null    int64  
 2   Pclass       418 non-null    int64  
 3   Name         418 non-null    object 
 4   Sex          418 non-null    object 
 5   Age          332 non-null    float64
 6   SibSp        418 non-null    int64  
 7   Parch        418 non-null    int64  
 8   Ticket       418 non-null    object 
 9   Fare         417 non-null    float64
 10  Cabin        91 non-null     object 
 11  Embarked     418 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
```

Out[7]:
```
Pclass
3    218
1    107
2     93
Name: count, dtype: int64
```

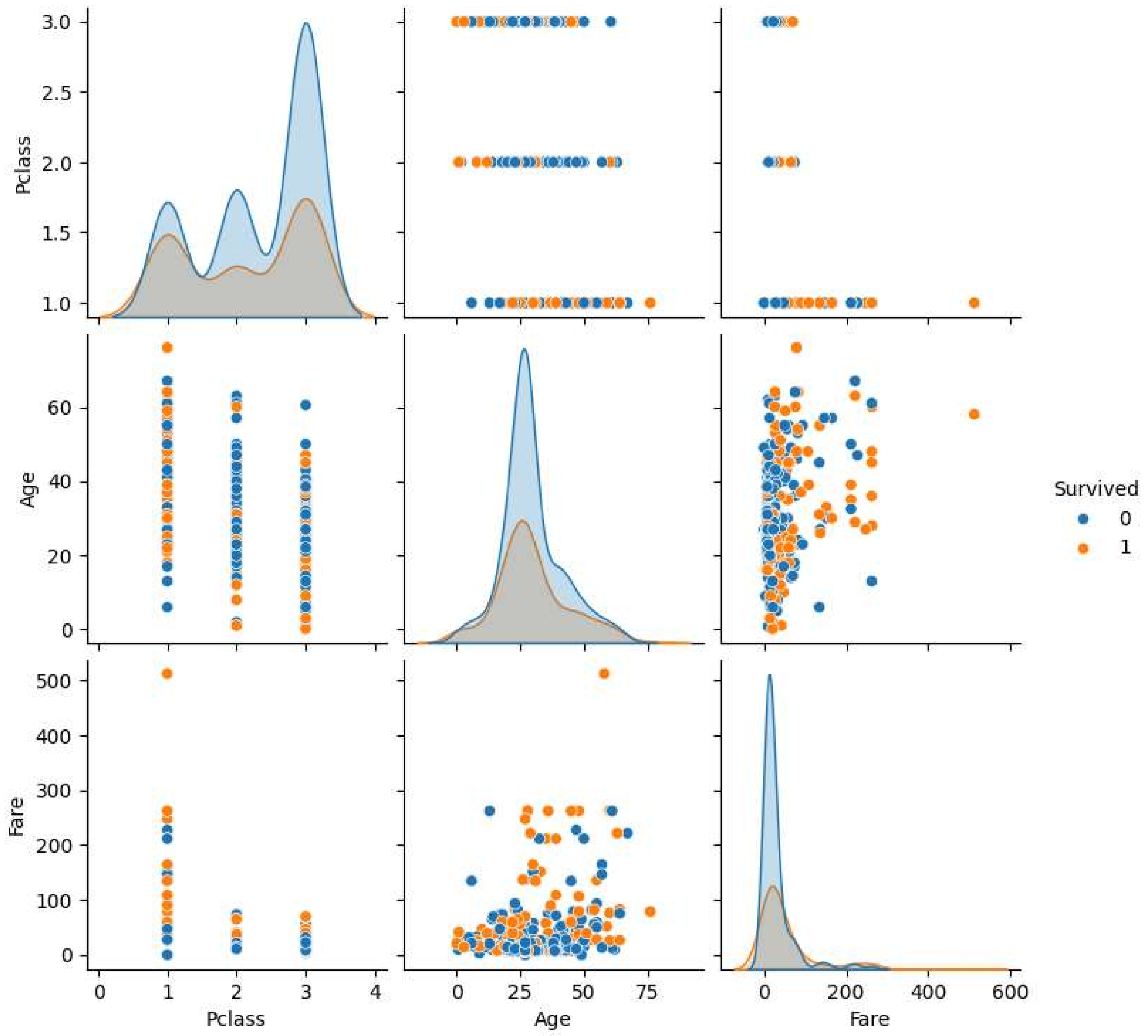
In [8]:
```
df.isnull().sum()
```

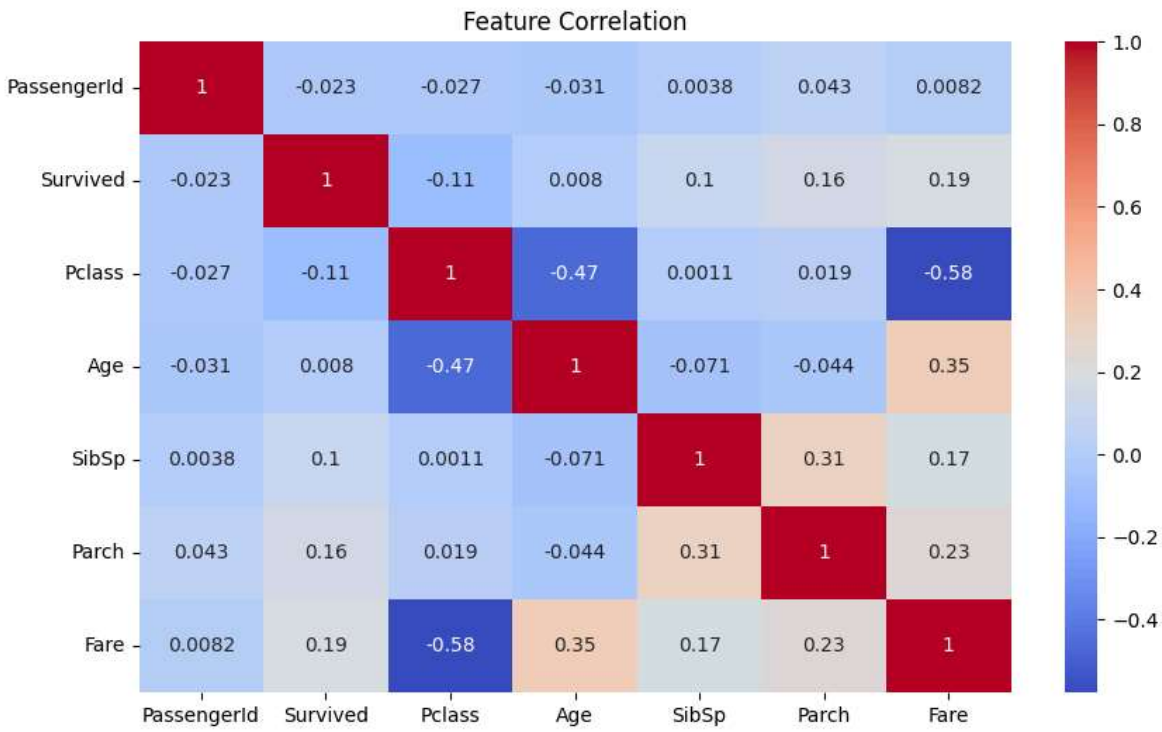
Out[8]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age             86
SibSp            0
Parch            0
Ticket           0
Fare             1
Cabin          327
Embarked         0
dtype: int64
```

In [9]:
```
df.drop(columns=['Cabin'], inplace=True)

df['Age'].fillna(df['Age'].median(), inplace=True)

df.dropna(subset=['Embarked'], inplace=True)
```

```
C:\Users\dhruv\AppData\Local\Temp\ipykernel_7548\3912125883.py:3: FutureWarning:
A value is trying to be set on a copy of a DataFrame or Series through chained as
signment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work becau
se the intermediate object on which we are setting values always behaves as a cop
y.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.meth
od({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to pe
rform the operation inplace on the original object.


  df['Age'].fillna(df['Age'].median(), inplace=True)
```

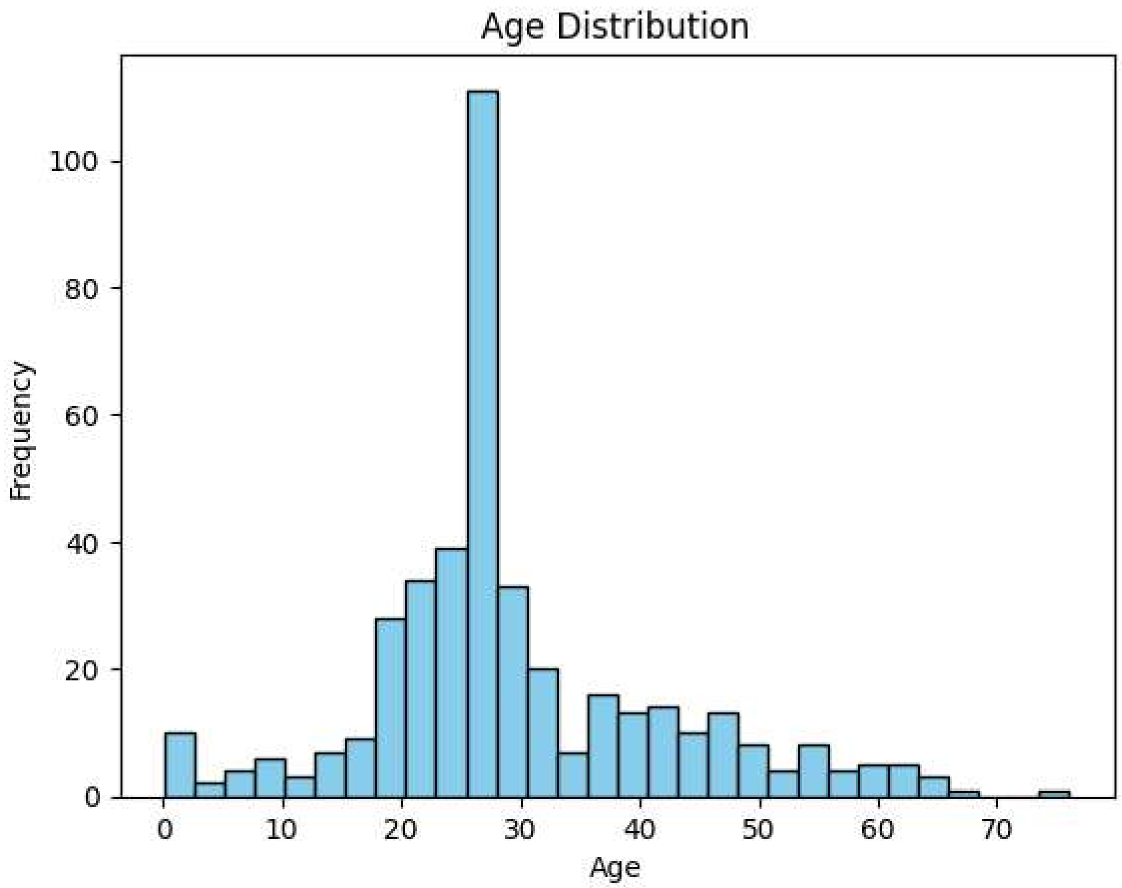
In [10]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Pairplot
sns.pairplot(df[['Survived', 'Pclass', 'Sex', 'Age', 'Fare']], hue='Survived')
plt.show()

# Heatmap of correlation
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Feature Correlation")
plt.show()
```

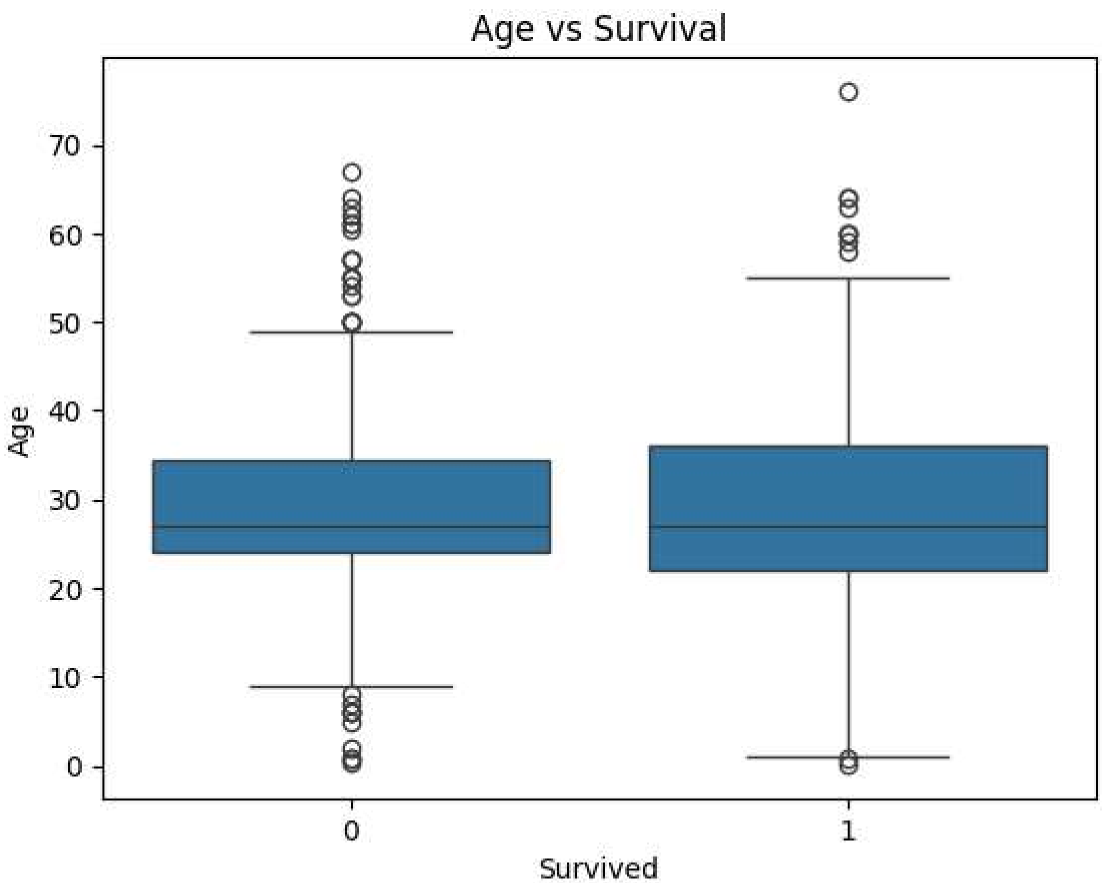


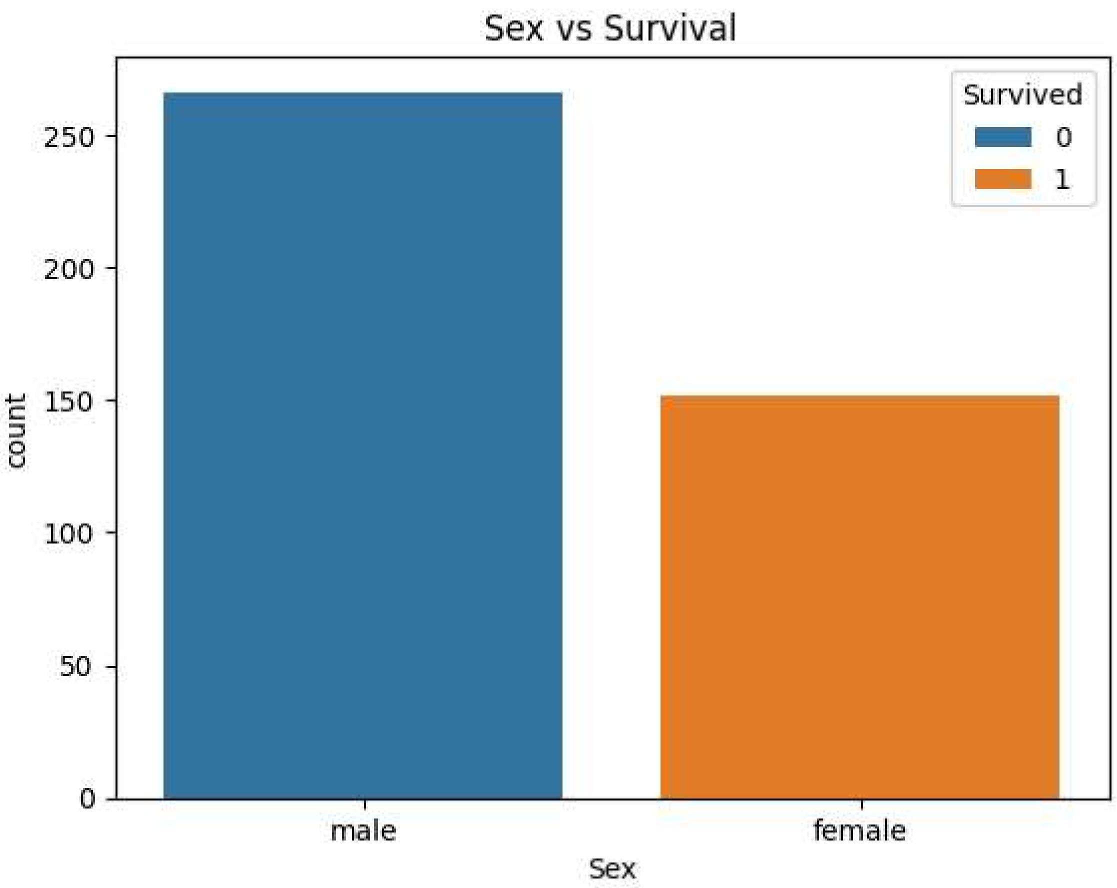

In [11]:
```
# Histogram of age
df['Age'].plot(kind='hist', bins=30, color='skyblue', edgecolor='black')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.show()
```

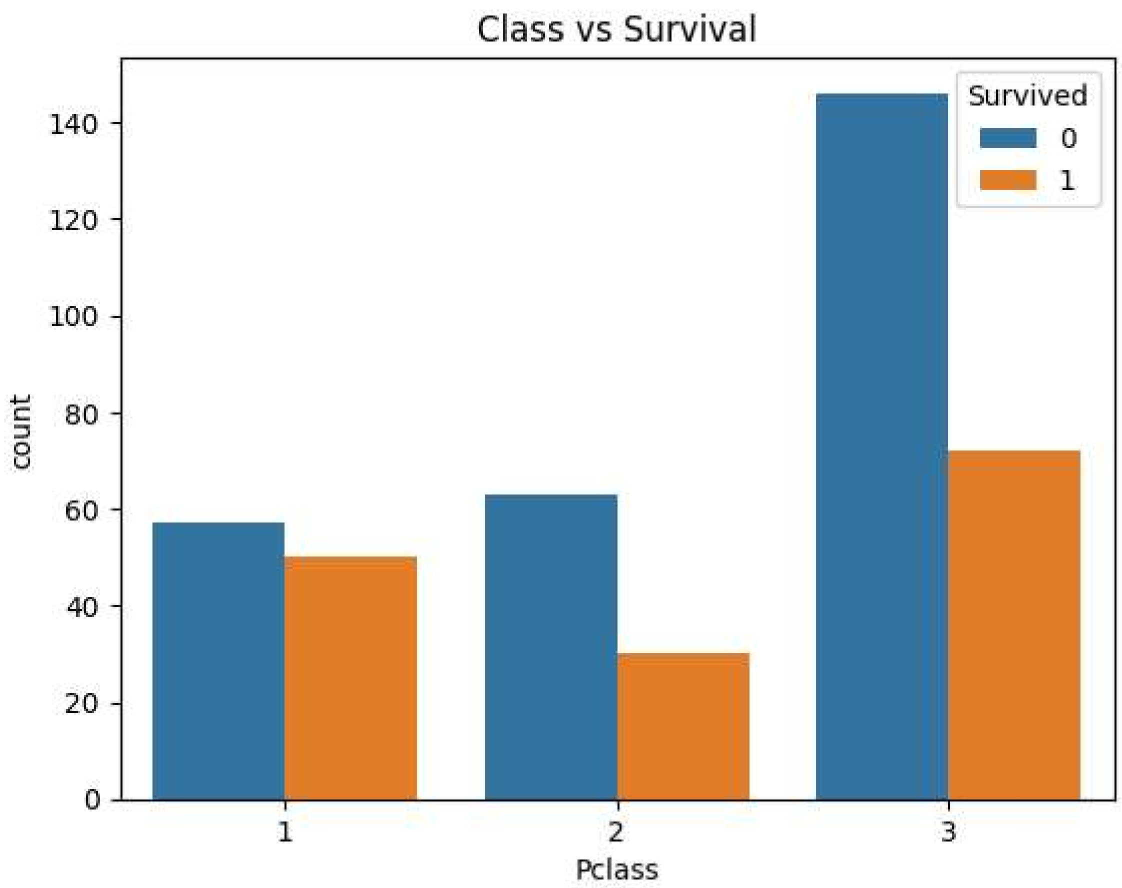
```
# Boxplot for age by survival
sns.boxplot(x='Survived', y='Age', data=df)
plt.title('Age vs Survival')
plt.show()
```



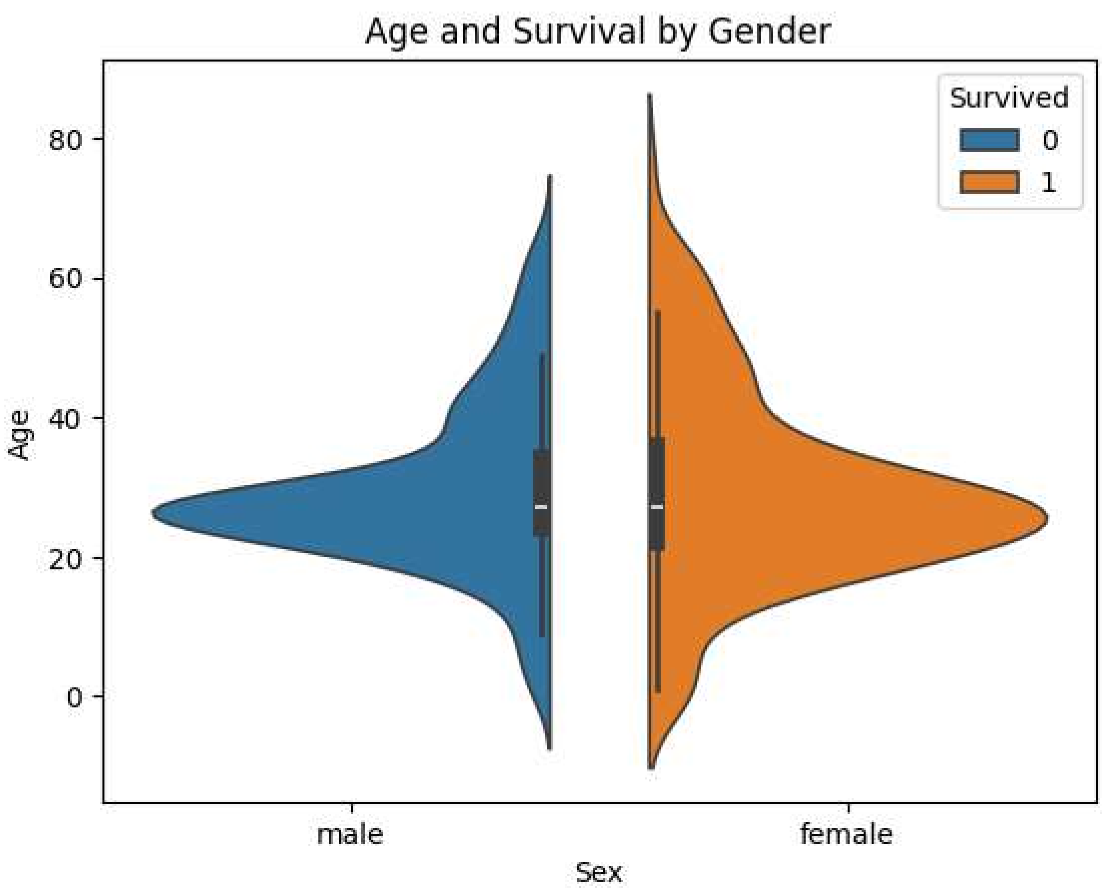



In [12]:
```
# Countplot for sex vs survival
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Sex vs Survival')
plt.show()

# Pclass vs survival
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Class vs Survival')
plt.show()
```






In [13]:
```
# Survival by age and sex
sns.violinplot(x='Sex', y='Age', hue='Survived', data=df, split=True)
plt.title('Age and Survival by Gender')
plt.show()
```



In [ ]: